

MM54C14/MM74C14 Hex Schmitt Trigger

General Description

The MM54C14/MM74C14 Hex Schmitt Trigger is a monolithic complementary MOS (CMOS) integrated circuit constructed with N- and P-channel enhancement transistors. The positive and negative going threshold voltages, V_{T+} and V_{T-} , show low variation with respect to temperature (typ. $0.0005V/^{\circ}C$ at $V_{CC} = 10V$), and hysteresis, $V_{T+} - V_{T-} \geq 0.2 V_{CC}$ is guaranteed.

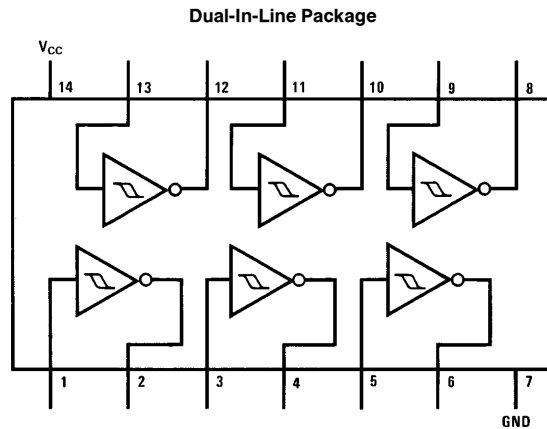
All inputs are protected from damage due to static discharge by diode clamps to V_{CC} and GND.

Features

- Wide supply voltage range
- High noise immunity
- Low power
- TTL compatibility
- Hysteresis

3.0V to 15V
 $0.70 V_{CC}$ (typ.)
 $0.4 V_{CC}$ (typ.)
 $0.2 V_{CC}$ guaranteed
 $0.4 V_{CC}$ (typ.)
 $0.2 V_{CC}$ guaranteed

Connection Diagram



TL/F/5879-1

Order Number MM54C14 or MM74C14

Absolute Maximum Ratings

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/Distributors for availability and specifications.

Voltage at Any Pin	−0.3V to V_{CC} + 0.3V
Operating Temperature Range	−55°C to +125°C
MM54C14	−40°C to +85°C
MM74C14	

Storage Temperature Range	−65°C to +150°C
Power Dissipation	
Dual-In-Line	700 mW
Small Outline	500 mW
Operating V_{CC} Range	3.0V to 15V
Absolute Maximum V_{CC}	18V
Lead Temperature	
(Soldering, 10 seconds)	260°C

DC Electrical Characteristics

Min/Max limits apply across the guaranteed temperature range unless otherwise noted

Symbol	Parameter	Conditions	Min	Typ	Max	Units
CMOS TO CMOS						
V_{T+}	Positive Going Threshold Voltage	$V_{CC} = 5V$	3.0	3.6	4.3	V
		$V_{CC} = 10V$	6.0	6.8	8.6	V
		$V_{CC} = 15V$	9.0	10.0	12.9	V
V_{T-}	Negative Going Threshold Voltage	$V_{CC} = 5V$	0.7	1.4	2.0	V
		$V_{CC} = 10V$	1.4	3.2	4.0	V
		$V_{CC} = 15V$	2.1	5.0	6.0	V
$V_{T+} - V_{T-}$	Hysteresis	$V_{CC} = 5V$	1.0	2.2	3.6	V
		$V_{CC} = 10V$	2.0	3.6	7.2	V
		$V_{CC} = 15V$	3.0	5.0	10.8	V
$V_{OUT(1)}$	Logical “1” Output Voltage	$V_{CC} = 5V, I_O = -10 \mu A$	4.5			V
		$V_{CC} = 10V, I_O = -10 \mu A$	9.0			V
$V_{OUT(0)}$	Logical “0” Output Voltage	$V_{CC} = 5V, I_O = 10 \mu A$			0.5	V
		$V_{CC} = 10V, I_O = 10 \mu A$			1.0	V
$I_{IN(1)}$	Logical “1” Input Current	$V_{CC} = 15V, V_{IN} = 15V$		0.005	1.0	μA
$I_{IN(0)}$	Logical “0” Input Current	$V_{CC} = 15V, V_{IN} = 0V$	-1.0	-0.005		μA
I_{CC}	Supply Current	$V_{CC} = 15V, V_{IN} = 0V/15V$		0.05	15	μA
		$V_{CC} = 5V, V_{IN} = 2.5V$ (Note 4)		20		μA
		$V_{CC} = 10V, V_{IN} = 5V$ (Note 4)		200		μA
		$V_{CC} = 15V, V_{IN} = 7.5V$ (Note 4)		600		μA
CMOS/LPTTL INTERFACE						
$V_{IN(1)}$	Logical “1” Input Voltage	$V_{CC} = 5V$	4.3			V
$V_{IN(0)}$	Logical “0” Input Voltage	$V_{CC} = 5V$			0.7	V
$V_{OUT(1)}$	Logical “1” Output Voltage	54C, $V_{CC} = 4.5V, I_O = -360 \mu A$	2.4			V
		74C, $V_{CC} = 4.75V, I_O = -360 \mu A$	2.4			V
$V_{OUT(0)}$	Logical “0” Output Voltage	54C, $V_{CC} = 4.5V, I_O = 360 \mu A$			0.4	V
		74C, $V_{CC} = 4.75V, I_O = 360 \mu A$			0.4	V
OUTPUT DRIVE (see 54C/74C Family Characteristics Data Sheet) $T_A = 25^{\circ}C$ (Short Circuit Current)						
I_{SOURCE}	Output Source Current (P-Channel)	$V_{CC} = 5V, V_{OUT} = 0V$	-1.75	-3.3		mA
I_{SOURCE}	Output Source Current (P-Channel)	$V_{CC} = 10V, V_{OUT} = 0V$	-8.0	-15		mA
I_{SINK}	Output Sink Current (N-Channel)	$V_{CC} = 5V, V_{OUT} = V_{CC}$	1.75	3.6		mA
I_{SINK}	Output Sink Current (N-Channel)	$V_{CC} = 10V, V_{OUT} = V_{CC}$	8.0	16		mA

AC Electrical Characteristics* $T_A = 25^\circ\text{C}$, $C_L = 50\text{ pF}$, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Units
t_{PD0} , t_{PD1}	Propagation Delay from Input to Output	$V_{CC} = 5\text{V}$		220	400	ns
		$V_{CC} = 10\text{V}$		80	200	ns
C_{IN}	Input Capacitance	Any Input (Note 2)		5.0		pF
C_{PD}	Power Dissipation Capacitance	(Note 3) Per Gate		20		pF

*AC Parameters are guaranteed by DC correlated testing.

Note 1: "Absolute Maximum Ratings" are those values beyond which the safety of the device cannot be guaranteed. Except for "Operating Temperature Range" they are not meant to imply that the devices should be operated at these limits. The table of "Electrical Characteristics" provides conditions for actual device operation.

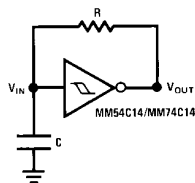
Note 2: Capacitance is guaranteed by periodic testing.

Note 3: C_{PD} determines the no load AC power consumption of any CMOS device. For complete explanation see 54C/74C Family Characteristics Application Note—AN-90.

Note 4: Only one of the six inputs is at $\frac{1}{2} V_{CC}$; the others are either at V_{CC} or GND.

Typical Applications

Low Power Oscillator



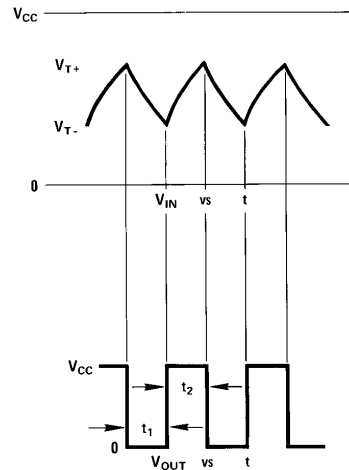
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$$t_1 \approx RC \ln \frac{V_{T+}}{V_{T-}}$$

$$t_1 \approx RC \ln \frac{V_{CC} - V_{T-}}{V_{CC} - V_{T+}}$$

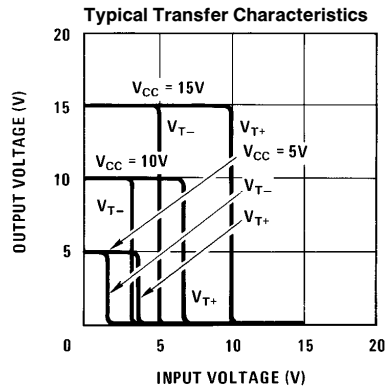
$$f \approx \frac{1}{RC \ln \frac{V_{T+}(V_{CC} - V_{T-})}{V_{T-}(V_{CC} - V_{T+})}} \approx \frac{1}{1.7 RC}$$

Note: The equations assume $t_1 + t_2 \gg t_{pd0} + t_{pd1}$

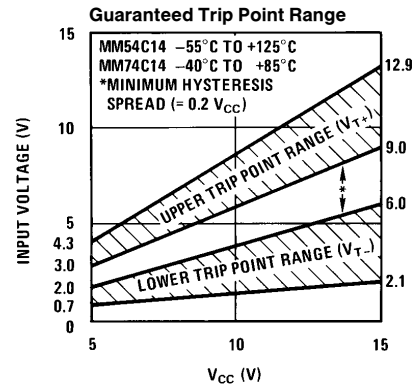


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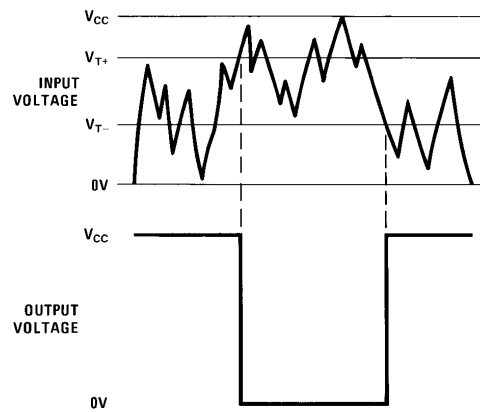
Typical Performance Characteristics



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TL/F/5879-5



TL/F/5879-6

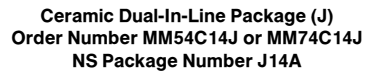
Note: For more information on output drive characteristics, power dissipation, and propagation delays, see AN-90.

Physical Dimensions inches (millimeters)

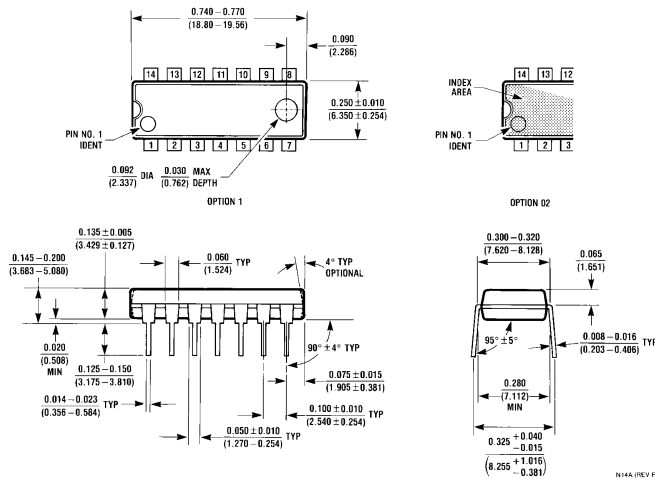
The diagram shows the physical dimensions of the J14A Ceramic Dual-In-Line Package (J) in inches and millimeters. The package is a rectangular component with 14 pins (7 on each side) and a central glass sealant area. Key dimensions include:

- Overall length: 0.785 (19.939) MAX
- Pin pitch: 0.025 (0.635) RAD
- Pin length: 0.220-0.310 (5.588-7.874)
- Pin diameter: 0.005 (0.127) MIN
- Pin spacing: 0.060 ± 0.005 (1.524 ± 0.127)
- Pin height: 0.018 ± 0.003 (0.457 ± 0.076)
- Pin base diameter: 0.100 ± 0.010 (2.540 ± 0.254)
- Pin base height: 0.150 (3.81) MIN
- Pin base width: 0.125-0.200 (3.175-5.080)
- Pin base thickness: 0.008-0.012 (0.203-0.305)
- Pin base angle: 86°-94° TYP
- Pin base radius: 0.098 (2.489) MAX BOTH ENDS
- Pin base width: 0.310-0.410 (7.874-10.41)
- Pin base height: 0.290-0.320 (7.366-8.128)
- Pin base thickness: 0.180 (4.572) MAX
- Pin base angle: 95° ± 5°
- Pin base width: 0.310-0.410 (7.874-10.41)
- Pin base height: 0.290-0.320 (7.366-8.128)
- Pin base thickness: 0.180 (4.572) MAX
- Pin base angle: 95° ± 5°
- Pin base width: 0.310-0.410 (7.874-10.41)
- Pin base height: 0.290-0.320 (7.366-8.128)
- Pin base thickness: 0.180 (4.572) MAX
- Pin base angle: 95° ± 5°

J14A (REV G)



Physical Dimensions inches (millimeters) (Continued)



Molded Dual-In-Line Package (N)
Order Number MM54C14N or MM74C14N
NS Package Number N14A

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